# 4. Decision Models

## 4.1 Decision Tree

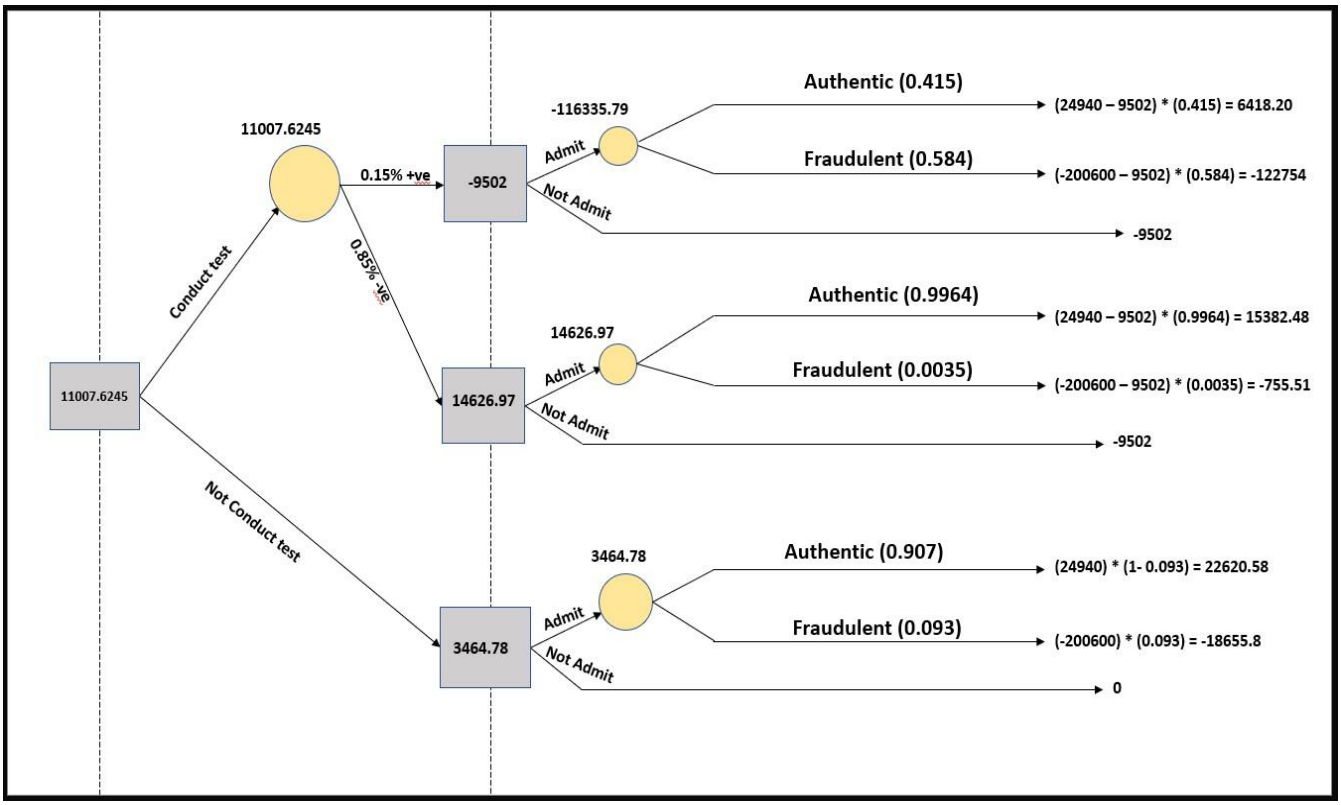


**Fig 4**

## 4.2 Expected Value of Sample Information

**EVSI** = **[Expected value with sample information + Cost of Sampling] – [Expected value of best decision without sample information]**

= (20509.6245 – 3964.78) **= 16,544.8445**

## 4.3 Efficiency of Sample Information

$$\textbf{Efficiency} = \frac{EVSI}{EVPI}$$

**Expected Value for Perfect Information (EVPI) = Expected Value with Perfect Information (EVwPI) – Maximum Expected Monetary Value**

**Maximum Expected Monetary Value (EMV) = EMV (A1) > EMV (A2)**

**EMV (A1)** = (24940*0.907) + (-200600*0.093)

= 3964.78

**EMV (A2)** = (0*0.907) + (0*0.093)

= 0.00

| | Authentic | Fraudulent |
|---|---|---|
| Admit | 24940 | -200600 |
| Not Admit | 0 | 0 |
| Probability | 0.907 | 0.093 |

**Table 12**

**Expected Value with Perfect Information (EVwPI) = (Best Payoff in State i) * (Probability of State i)**

= (24940*0.907) + (0*0.093)

= 22620.58

**EVPI =** 22620.58 – 3964.78

= 18655.8

**Efficiency =** $\frac{EVSI}{EVPI}$

$= \frac{16544.8445}{18655.8}$

= 0.8868 / 88.00 %

Therefore, Efficiency of sample Information is 88.00%